

AAA Options:

- o Cisco provides a number of ways to implement AAA.
- o Two main protocols are TACACS Server and Radius Server.

AAA with TACACS+:

- o TACACS+ stands for Terminal Access Controller Access Control System Plus.
- o TACACS+ is a Cisco proprietary protocol that is use to deliver AAA security services.
- o TACACS+ is an application, which is implement through AAA.
- o TACACS+ provides centralized acceptance of user to take the access control of routers.
- o TACACS+ provides other access servers in the network as well.
- o TACACS+ provides to control the authorization of router commands per-user or group.
- o TACACS+ offers multiprotocol support.
- o TACACS+ encrypts entire body of the packet but leaves a standard TACACS+ header.
- o TACACS+ uses the AAA architecture, which separates AAA.

AAA with RADIUS:

- o RADIUS stand for Remote Authentication Dial in User Service.
- o RADIUS is a security protocol that secures the network against unauthorized access.
- o RADIUS clients run on routers & send authentication request to a centralized server.
- o RADIUS Server contains network service access information and user authentication.
- o RADIUS does not allow users to control which commands can be executed or not.
- o RADIUS is not as useful for router or switch management.
- o RADIUS does not support multiprotocol.
- o RADIUS encrypts password of the access-request packet only from Client to server.
- o RADIUS uses UDP as a transport protocol while TACACS+ uses TCP.
- o RADIUS combines authentication and authorization processes.

RADIUS	TACACS+
RADIUS uses UDP	TACACS+ uses TCP
Uses ports 1812/1645 for authentication Uses ports 1813/1646 for accounting	TACACS+ uses TCP port 49
RADIUS encrypts passwords only	TACACS+ encrypts the entire communication
RADIUS combines authentication and Authorization	TACACS+ treats Authentication, Authorization, and Accountability differently
RADIUS is an open protocol	TACACS+ is Cisco proprietary protocol
RADIUS is a light-weight protocol consuming less resources	TACACS+ is a heavy-weight protocol consuming more resources
RADIUS is limited to privilege mode	TACACS+ supports 15 privilege levels
Mainly used for Network Access	Mainly used for Device Administration